

# Linear Test 5 Math

**The SAT<sup>®</sup>**

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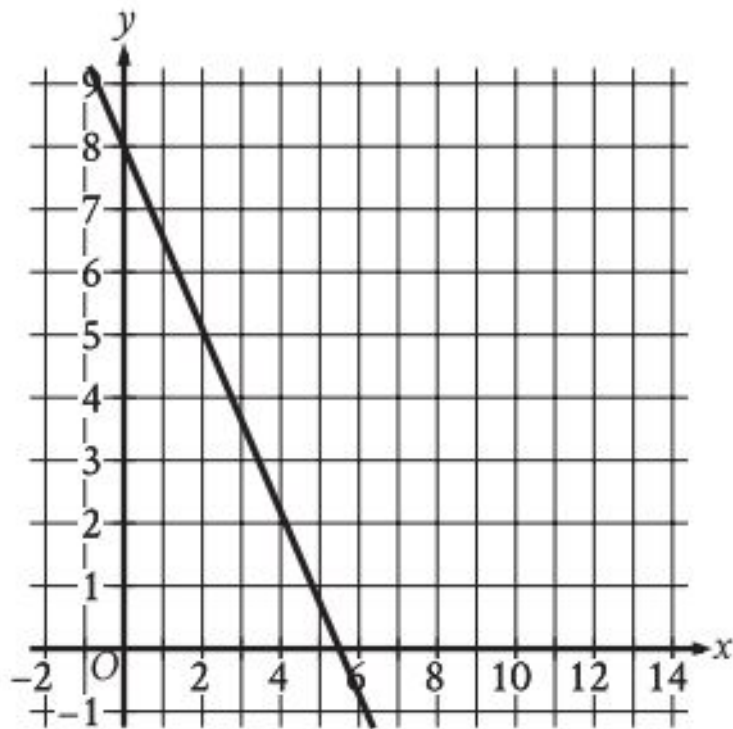
Practice

Test #5



On the first day of a semester, a film club has 90 members. Each day after the first day of the semester, 10 new members join the film club. If no members leave the film club, how many total members will the film club have 4 days after the first day of the semester?

- A) 400
- B) 130
- C) 94
- D) 90



A)  $(0, 0)$

B)  $\left(0, -\frac{16}{11}\right)$

C)  $(0, -8)$

D)  $(0, 8)$

The graph of the linear function  $f$  is shown, where  $y = f(x)$ . What is the  $y$ -intercept of the graph of  $f$ ?

$$\begin{aligned}s + 7r &= 27 \\ r &= 3\end{aligned}$$

What is the solution  $(r, s)$  to the given system of equations?

- A)  $(6, 3)$
- B)  $(3, 6)$
- C)  $(3, 27)$
- D)  $(27, 3)$

The table shows selected values from function  $f$ .

| $x$ | $f(x)$ |
|-----|--------|
| -1  | 16     |
| 0   | 17     |
| 1   | 18     |
| 2   | 19     |

Which of the following is the best description of function  $f$ ?

- A) Decreasing linear
- B) Increasing linear
- C) Decreasing exponential
- D) Increasing exponential

Which of the following functions has(have) a minimum value at  $-3$  ?

I.  $f(x) = -6(3)^x - 3$

II.  $g(x) = -3(6)^x$

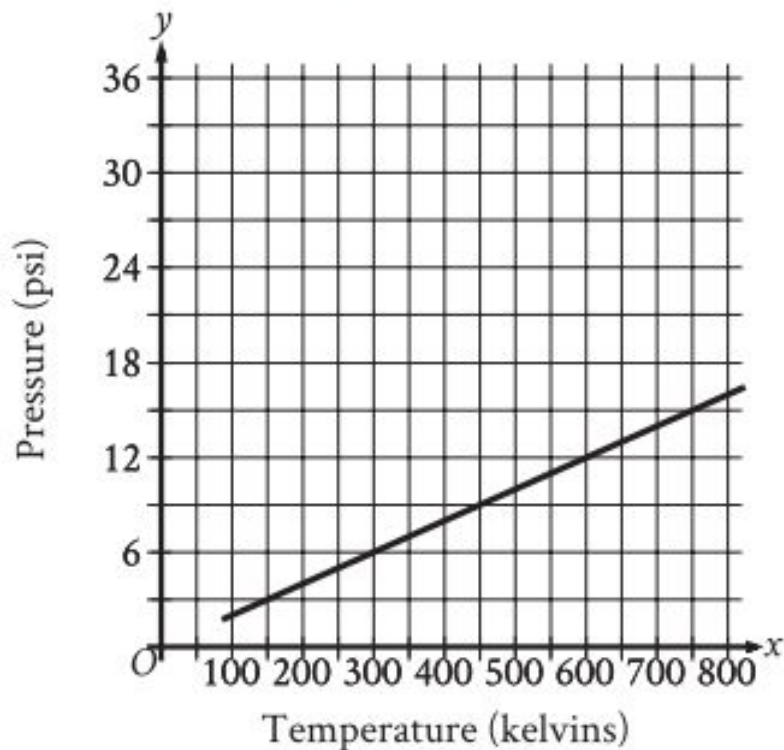
A) I only

B) II only

C) I and II

D) Neither I nor II

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure  $y$ , in pounds per square inch (psi), of the argon when its temperature is  $x$  kelvins.



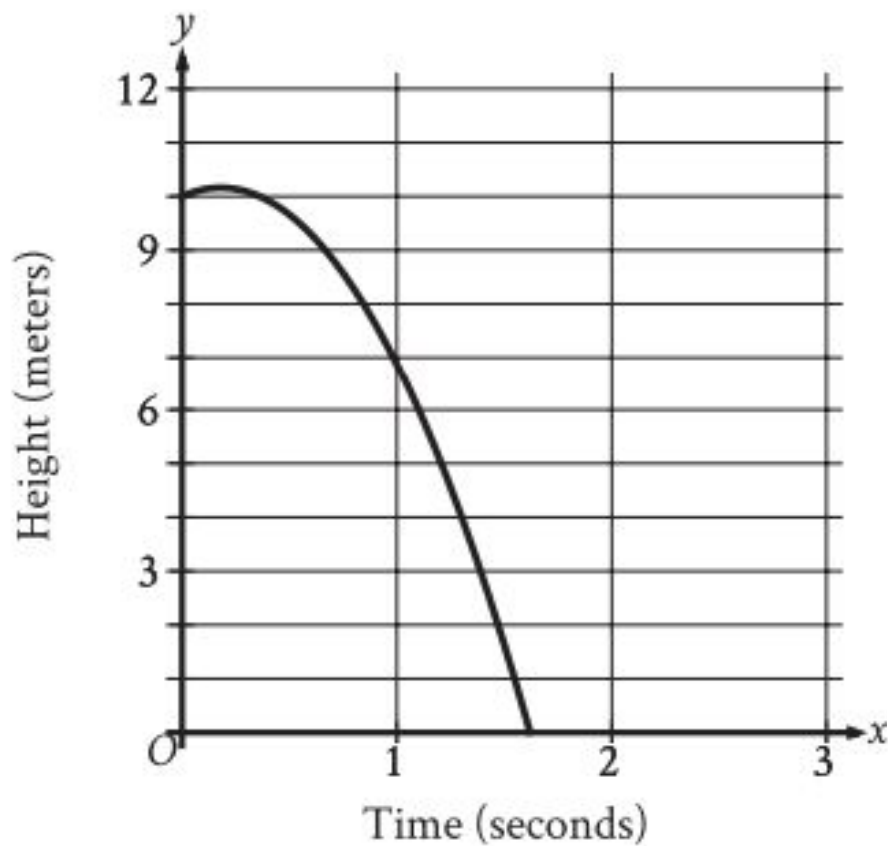
What is the estimated pressure of the argon, in psi, when the temperature is 600 kelvins?

- A) 6
- B) 12
- C) 300
- D) 600

Which expression is equivalent to  $16x^3y^2 + 14xy$  ?

- A)  $2xy(8xy + 7)$
- B)  $2xy(8x^2y + 7)$
- C)  $14xy(2x^2y + 1)$
- D)  $14xy(8x^2y + 1)$





A competitive diver dives from a platform into the water. The graph shown gives the height above the water  $y$ , in meters, of the diver  $x$  seconds after diving from the platform. What is the best interpretation of the  $x$ -intercept of the graph?

- A) The diver reaches a maximum height above the water at 1.6 seconds.
- B) The diver hits the water at 1.6 seconds.
- C) The diver reaches a maximum height above the water at 0.2 seconds.
- D) The diver hits the water at 0.2 seconds.

The kinetic energy, in joules, of an object with mass 9 kilograms traveling at a speed of  $v$  meters per second is given by the function  $K$ , where

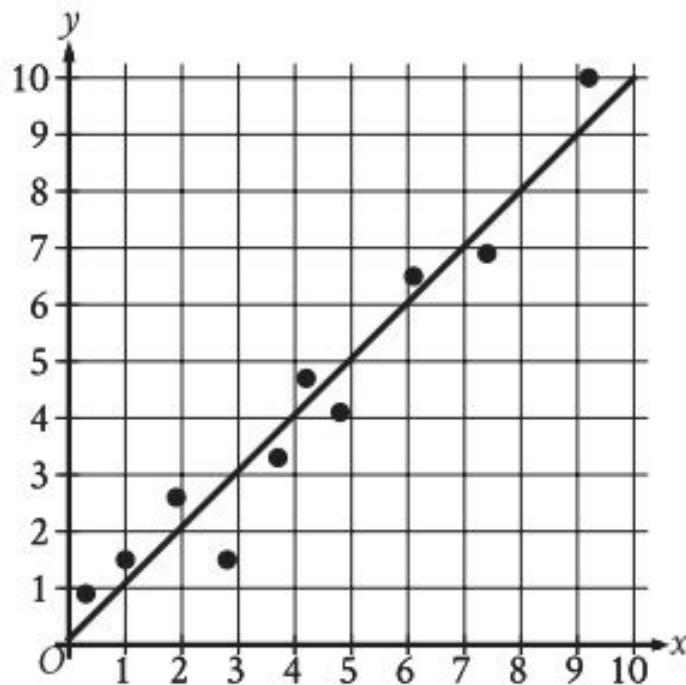
$$K(v) = \frac{9}{2} v^2.$$

Which of the following is the best

interpretation of  $K(34) = 5,202$  in this context?

- A) The object traveling at 34 meters per second has a kinetic energy of 5,202 joules.
- B) The object traveling at 340 meters per second has a kinetic energy of 5,202 joules.
- C) The object traveling at 5,202 meters per second has a kinetic energy of 34 joules.
- D) The object traveling at 23,409 meters per second has a kinetic energy of 34 joules.

The scatterplot shows the relationship between two variables  $x$  and  $y$ . A line of best fit for the data is also shown.



For how many of the 10 data points is the actual  $y$ -value greater than the  $y$ -value predicted by the line of best fit?

- A) 3
- B) 4
- C) 6
- D) 7

$$5y = 10x + 11$$

$$-5y = 5x - 21$$

The solution to the given system of equations is  $(x, y)$ . What is the value of  $30x$  ?

Rectangles  $ABCD$  and  $EFGH$  are similar. The length of each side of  $EFGH$  is 6 times the length of the corresponding side of  $ABCD$ . The area of  $ABCD$  is 54 square units. What is the area, in square units, of  $EFGH$ ?

- A) 9
- B) 36
- C) 324
- D) 1,944

$$P(t) = 260(1.04)^{\left(\frac{6}{4}\right)t}$$

The function  $P$  models the population, in thousands, of a certain city  $t$  years after 2003. According to the model, the population is predicted to increase by 4% every  $n$  months. What is the value of  $n$  ?

- A) 8
- B) 12
- C) 18
- D) 72

A circle in the  $xy$ -plane has its center at  $(-1, 1)$ .  
Line  $t$  is tangent to this circle at the point  $(5, -4)$ .  
Which of the following points also lies on line  $t$  ?

A)  $\left(0, \frac{6}{5}\right)$

B)  $(4, 7)$

C)  $(10, 2)$

D)  $(11, 1)$